

QUESTION 1

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, calinski_harabasz_score

# Load dataset
df = pd.read_csv('Dataset_Day13.csv')

# Check for missing values
print(df.isnull().sum())

# fill or drop missing values
df.dropna(inplace=True)

# Detect and treat outliers (using IQR method)
numeric_cols = ['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']

for col in numeric_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    df = df[(df[col] >= lower_bound) & (df[col] <= upper_bound)]
print("After cleaning, dataset shape:", df.shape)
```

```
Id          0
SepalLengthCm  0
SepalWidthCm  0
PetalLengthCm  0
PetalWidthCm  0
Species      0
dtype: int64
After cleaning, dataset shape: (146, 6)
```

QUESTION 2

```
X = df[numeric_cols]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Group by species and calculate mean petal length
avg_petal_length = df.groupby('Species')['PetalLengthCm'].mean()
print(avg_petal_length)

# Only numeric columns grouped by Species
desc_stats = df.groupby('Species')[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']].describe()
print(desc_stats)
```

```
After cleaning, dataset shape: (146)
Species
Iris-setosa      1.46383
Iris-versicolor  4.27551
Iris-virginica   5.55200
Name: PetalLengthCm, dtype: float64
```

iris-virginica 5155200

Name: PetalLengthCm, dtype: float64

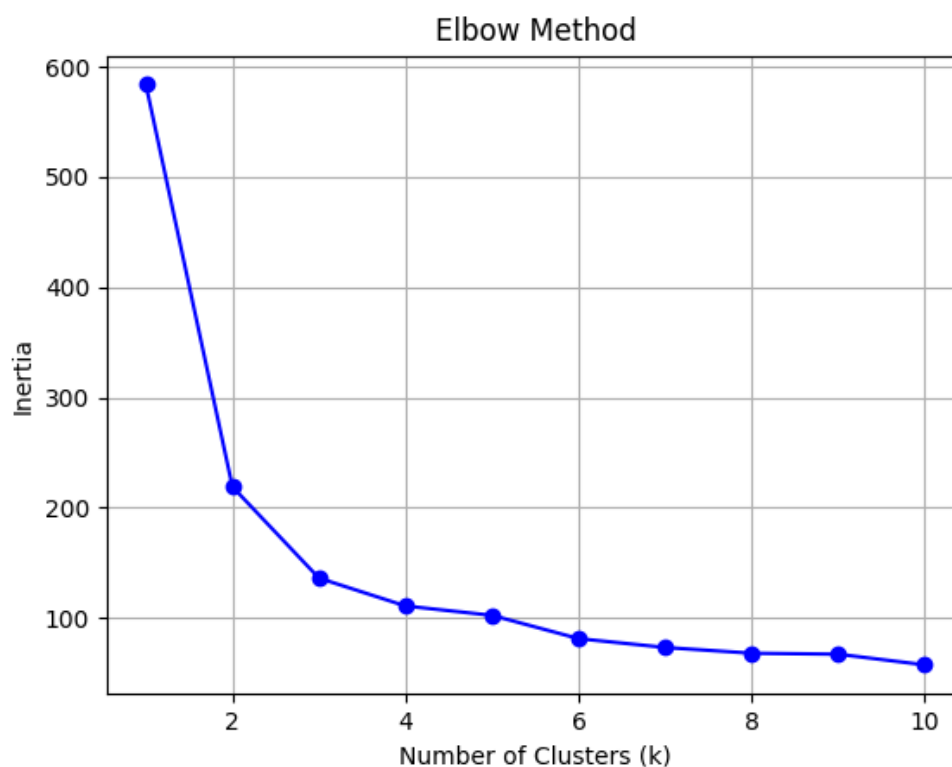
		SepalLengthCm				... PetalWidthCm						
	count	mean	std	min	25%	...	min	25%	50%	75%	max	
Species						...						
Iris-setosa	47.0	4.976596	0.338932	4.3	4.800	...	0.1	0.2	0.2	0.3	0.6	
Iris-versicolor	49.0	5.955102	0.503348	4.9	5.600	...	1.0	1.2	1.3	1.5	1.8	
Iris-virginica	50.0	6.588000	0.635880	4.9	6.225	...	1.4	1.8	2.0	2.3	2.5	

QUESTION 4

```
# Elbow Method
inertias = []
K = range(1, 11)
for k in K:
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertias.append(kmeans.inertia_)

plt.plot(K, inertias, 'bo-')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.grid(True)
plt.show()

# Silhouette and Calinski-Harabasz Scores for odd k
print("\nSilhouette and Calinski-Harabasz Scores:")
for k in [3, 5, 7, 9]:
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(X_scaled)
    sil = silhouette_score(X_scaled, labels)
    ch = calinski_harabasz_score(X_scaled, labels)
    print(f"k = {k} | Silhouette Score = {sil:.3f} | Calinski-Harabasz Score = {ch:.2f}")
```



Silhouette and Calinski-Harabasz Scores:

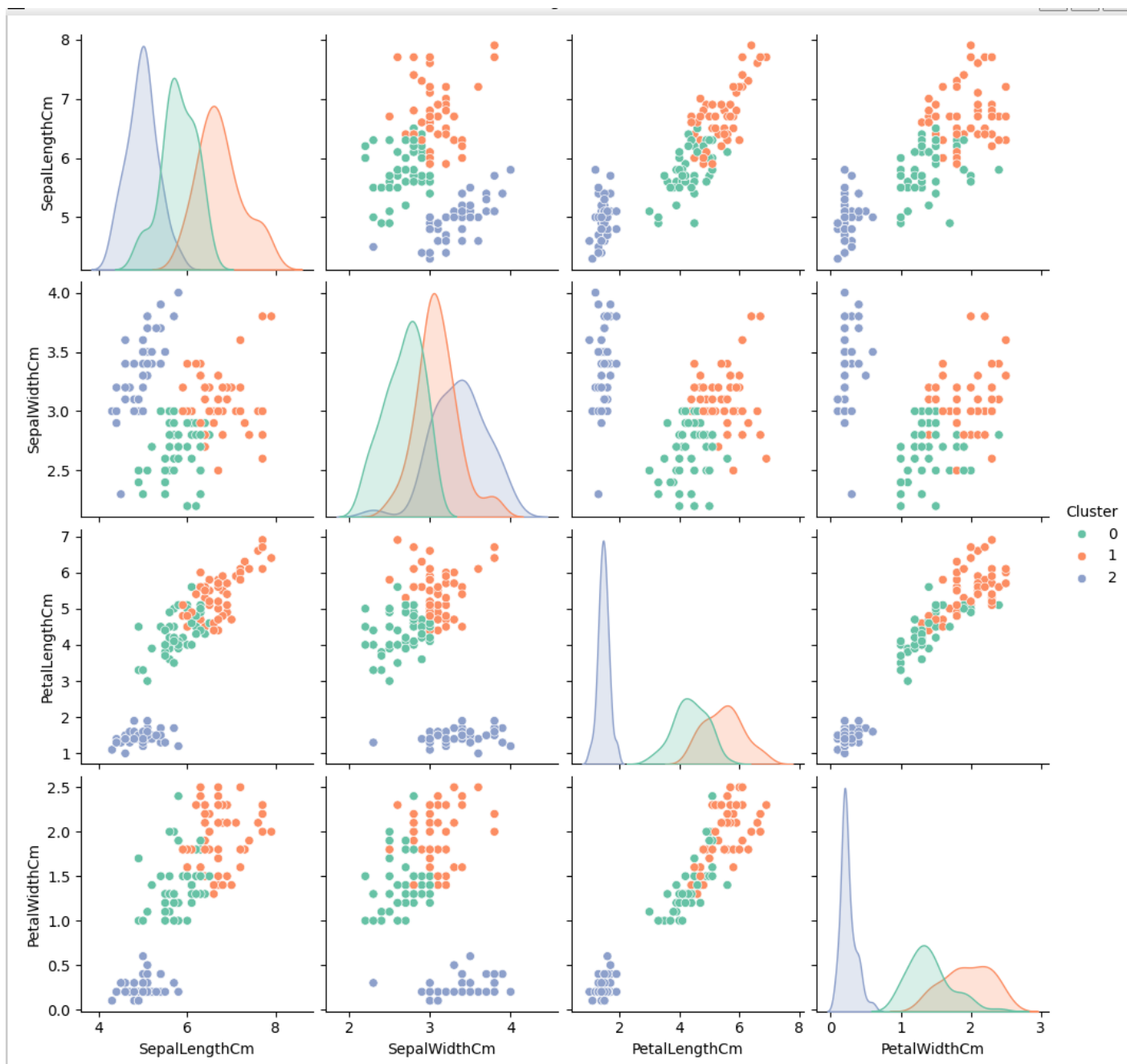
k = 3	Silhouette Score = 0.457	Calinski-Harabasz Score = 235.37
k = 5	Silhouette Score = 0.368	Calinski-Harabasz Score = 166.05
k = 7	Silhouette Score = 0.299	Calinski-Harabasz Score = 161.84
k = 9	Silhouette Score = 0.314	Calinski-Harabasz Score = 132.12

Question 5 AND 3

```
optimal_k = 3
kmeans = KMeans(n_clusters=optimal_k, random_state=42)
df['Cluster'] = kmeans.fit_predict(X_scaled)

proportion_table = pd.crosstab(df['Cluster'], df['Species'], normalize='index')
print("\n Proportion of Species in Each Cluster:")
print(proportion_table)

sns.pairplot(df, hue='Cluster', vars=numeric_cols, palette='Set2')
plt.suptitle("K-Means Cluster Visualization", y=1.02)
plt.show()
```



Proportion of Species in Each Cluster:			
Species	Iris-setosa	Iris-versicolor	Iris-virginica
Cluster			
0	0.0	0.755102	0.244898
1	0.0	0.240000	0.760000
2	1.0	0.000000	0.000000

Question 6

```
print("\n Insights based on Clustering:")

if proportion_table.shape[0] >= 3:
    cluster_0 = proportion_table.iloc[0].idxmax()
    cluster_1 = proportion_table.iloc[1].idxmax()
    cluster_2 = proportion_table.iloc[2].idxmax()
    print(f"- Cluster 0 mostly contains: {cluster_0}")
    print(f"- Cluster 1 mostly contains: {cluster_1}")
    print(f"- Cluster 2 mostly contains: {cluster_2}")

print("""
- The Elbow method shows a sharp drop at k = 3, supporting it as the optimal number of clusters.
- Silhouette and Calinski-Harabasz scores also peak or are relatively high at k = 3.
- Cluster 0 is almost purely Iris-setosa, indicating it is linearly separable from the others.
- Versicolor and Virginica are mixed in Cluster 1 and 2 – these species are not linearly separable.
- K-Means was effective at separating Setosa but not perfect for the other two.
""")
```

Insights based on Clustering:

- Cluster 0 mostly contains: Iris-versicolor
 - Cluster 1 mostly contains: Iris-virginica
 - Cluster 2 mostly contains: Iris-setosa
-
- The Elbow method shows a sharp drop at k = 3, supporting it as the optimal number of clusters.
 - Silhouette and Calinski-Harabasz scores also peak or are relatively high at k = 3.
 - Cluster 0 is almost purely Iris-setosa, indicating it is linearly separable from the others.
 - Versicolor and Virginica are mixed in Cluster 1 and 2 – these species are not linearly separable.
 - K-Means was effective at separating Setosa but not perfect for the other two.